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Remote Tech Support

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This project aims to build and implement a relational database for the Remote Tech Support Services system. The database organizes customer information, device details, support tickets, technician activities, and technical problem resolution. To provide remote technical support, it's important to maintain an orderly system for tracking client requests, device issues, technicians, and troubleshooting efforts. A structured database can help manage and access information more easily. This project's database stores and connects customer, device, technician, support, and ticket information.

Requirements Gathering and System Design

The requirements were mostly based on my own experience as a Linux system administrator. I gained expertise troubleshooting technical issues, providing support to users and systems, and documenting problem resolution processes. To determine what information should be stored in the database, I followed a typical support workflow: a client reports a device issue, a support ticket is created to record the problem and status, a technician handles the ticket, and ticket logs describe troubleshooting steps.

Key requirements:

- Store customer contact information as well as their devices.
- Maintain technician information, including specialization and tier level.
- Monitor support tickets, affected devices, issue descriptions, status, creation, and resolution times.
- Associate technicians with the tickets they manage.
- Enable several troubleshooting log entries for a ticket, such as date, action type, and notes.

Entities & Relationships

Five entities were created: Customer, Device, Technician, Support Ticket, and Ticket Log. Customer stores contact information; device stores customer device information; technician stores support staff information; Support Ticket stores reported technical issues, and Ticket Log records troubleshooting activity. Primary keys uniquely identify each record, while foreign keys connect related records. The main one-to-many relationships are Customer-to-Device, Customer-to-Support Ticket, Device-to-Support Ticket, Technician-to-Support Ticket, and Support Ticket-to-Ticket Log

ER Diagram Design

Two ER diagrams represent the system. The hand-drawn Chen-style diagram shows the entities, attributes, relationships, and cardinalities conceptually. The UML-style EER diagram was created in MySQL Workbench and represents the implemented tables, columns, primary keys, foreign keys, and relationships.

Database Implementation & SQL

The database was created in MySQL Workbench as `remote_tech_support` with five tables: `customer`, `devices`, `technician`, `supportTickets`, and `ticketLogs`. The Workbench model was forward engineered to generate the database structure. SQL scripts were used to insert sample customers, technicians, devices, tickets, and ticket logs. `SELECT` statements verified the data, and a `JOIN` query combined support ticket, customer, and technician information to demonstrate retrieval across related tables.

Challenges

During Forward Engineering, `technicianID` was initially configured to allow `NULL` while also serving as a primary key. This caused an error because a primary key cannot contain `NULL` values. I corrected the column to `NOT NULL` and verified the primary key settings. I also accidentally executed an `INSERT` statement multiple times, which created duplicate technician records. Correcting the duplicates reinforced the importance of carefully selecting statements before executing SQL in MySQL Workbench.

Outcome & Conclusion

The completed database provides a working relational structure for managing customers, devices, technicians, support tickets, and troubleshooting logs. The project helped me apply requirements gathering, Chen and UML ER modeling, primary and foreign keys, SQL, JOIN operations, Forward Engineering, and database troubleshooting. It also demonstrated how a real technical-support workflow can be translated into an organized relational database that makes support information easier to maintain and retrieve.